Research Initiative: Peak and Off-Peak Traffic Patterns in the Boroondara SCATS Dataset

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1. Purpose of the Research Initiative

The purpose of this research initiative was to investigate whether the traffic data provided for the project contains clear patterns that justify the need for traffic-aware route guidance. The main system is intended to predict traffic flow and use those predictions to estimate travel time. However, before those predictions can be meaningful, it is useful to establish whether traffic conditions actually vary across time and location in the supplied Boroondara dataset.

This analysis therefore focuses on two practical questions: when does traffic become most congested, and which recorded locations carry the highest volume? These questions are closely related to the final TBRGS because a route that is efficient during low-flow conditions may not remain efficient during the evening peak.

2. Research Question

How does traffic flow vary across different times of the day and SCATS locations in the Boroondara dataset, and what does this indicate about the importance of traffic-aware route guidance?

3. Dataset and Method

The analysis used the supplied file, Scats Data October 2006.xls, specifically the Data worksheet. Each daily record contains a SCATS number, recorded location, date and 96 traffic-volume values (V00 to V95). These values represent vehicle counts measured at 15-minute intervals over a full day.

The data was inspected for completeness and duplication before analysis. The 96 volume fields were then reshaped into a time-based format so that average traffic flow could be calculated for each 15-minute interval. Further summaries were generated by traffic period, location and day type (weekday or weekend). The analysis was descriptive: it does not replace the compulsory machine-learning evaluation, but adds context for understanding where traffic prediction is most useful.

Table 1. Summary of the supplied dataset

Measure	Result
Daily traffic records	4,192
Unique SCATS sites	40
Recorded locations/directions	139
Study period	1 October to 31 October 2006
Traffic readings per day	96
Measurement interval	15 minutes
Missing traffic-volume values	0
Duplicate records	0

The initial quality check found no missing traffic-volume values and no duplicated daily records. This gives a sound basis for analysing daily traffic patterns, because the observed peaks are not caused by obvious gaps or duplicate observations in the volume fields.

4. Traffic Variation Across the Day

The average traffic profile displays a strong time-of-day pattern. Flow remains very low during the early morning, increases rapidly from around 6:00 am, and reaches a clear morning commuting peak. Traffic remains relatively high through the day before reaching its largest peak in the late afternoon and early evening.

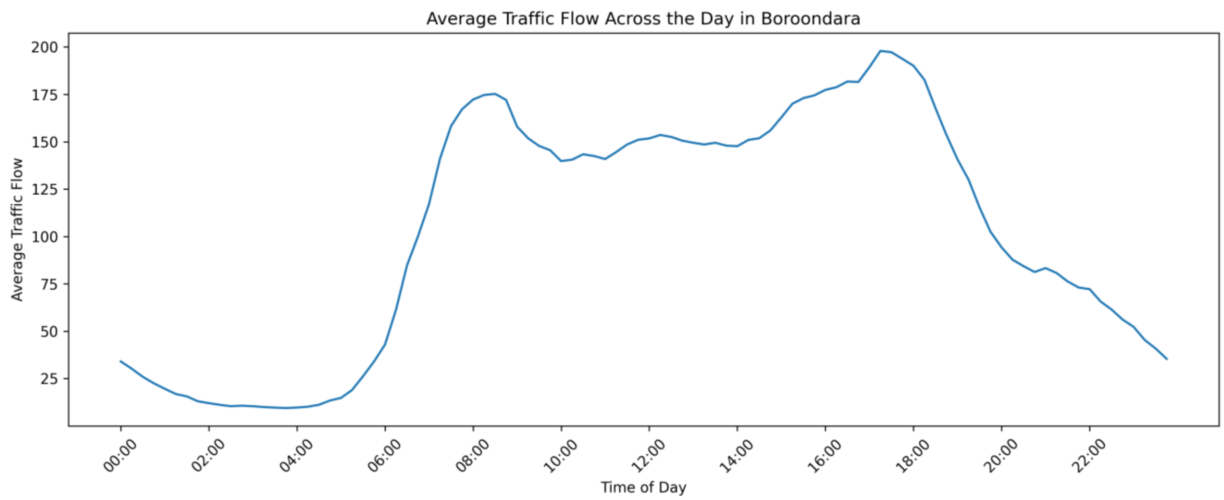

Figure 1. Average traffic flow across 96 fifteen-minute intervals in the Boroondara dataset.

Table 2. Five highest average traffic intervals

Time Interval	Average Flow	Maximum Observed Flow
17:15	197.93	617
17:30	197.25	608
17:45	193.71	636
18:00	190.11	629
17:00	189.37	593

The busiest average interval was 5:15 pm, with an average traffic flow of 197.93. The next highest intervals also occurred during the same evening period, confirming that the largest concentration of traffic in this dataset occurs around the end of the working day. At the other extreme, the quietest interval was 3:45 am, with an average flow of only 9.44. This sharp difference shows that traffic volume cannot be treated as a constant cost throughout the day.

5. Peak and Off-Peak Comparison

To make the daily pattern easier to interpret, selected intervals were grouped into broader periods. The resulting comparison shows that the evening peak carries the highest typical traffic volume, while overnight traffic is substantially lower.

Table 3. Average flow across selected daily periods

Period	Time Range	Average Flow	Maximum	Median
Overnight	00:00–05:45	16.66	304	11
Morning peak	07:00–08:45	159.75	695	138
Midday	10:00–14:45	148.07	620	141
Evening peak	16:00–17:45	187.22	636	176
Late evening	20:00–23:45	68.13	391	62

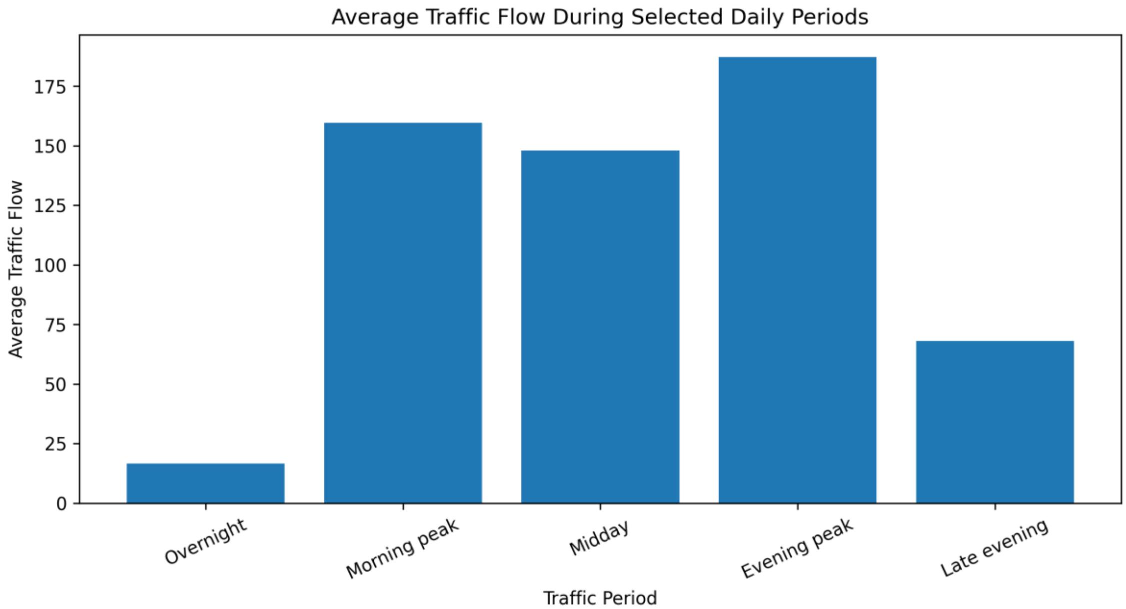

Figure 2. Comparison of average traffic flow during selected daily periods.

The evening peak recorded an average flow of 187.22, compared with only 16.66 during the overnight period. In practical terms, average flow during the evening peak was more than eleven times the overnight level. For route guidance, this matters because a road segment that is manageable overnight may form part of a much slower route during the evening commute.

6. Locations with the Highest Traffic Flow

The analysis also identified the recorded locations with the highest average flow. This is useful because a traffic-aware routing system should be particularly sensitive to high-volume road sections, where a small deterioration in conditions may affect total journey time.

Table 4. Five busiest recorded locations by average traffic flow

SCATS Site	Recorded Location	Average Flow	Maximum
3685	WARRIGAL RD S of HIGHBURY RD	261.49	636
3001	BARKERS RD W of CHURCH ST	191.78	620
2000	WARRIGAL RD S of BURWOOD HWY	189.15	464
970	WARRIGAL RD N of HIGH STREET RD	188.85	450
3685	WARRIGAL RD N of HIGHBURY RD	185.11	397

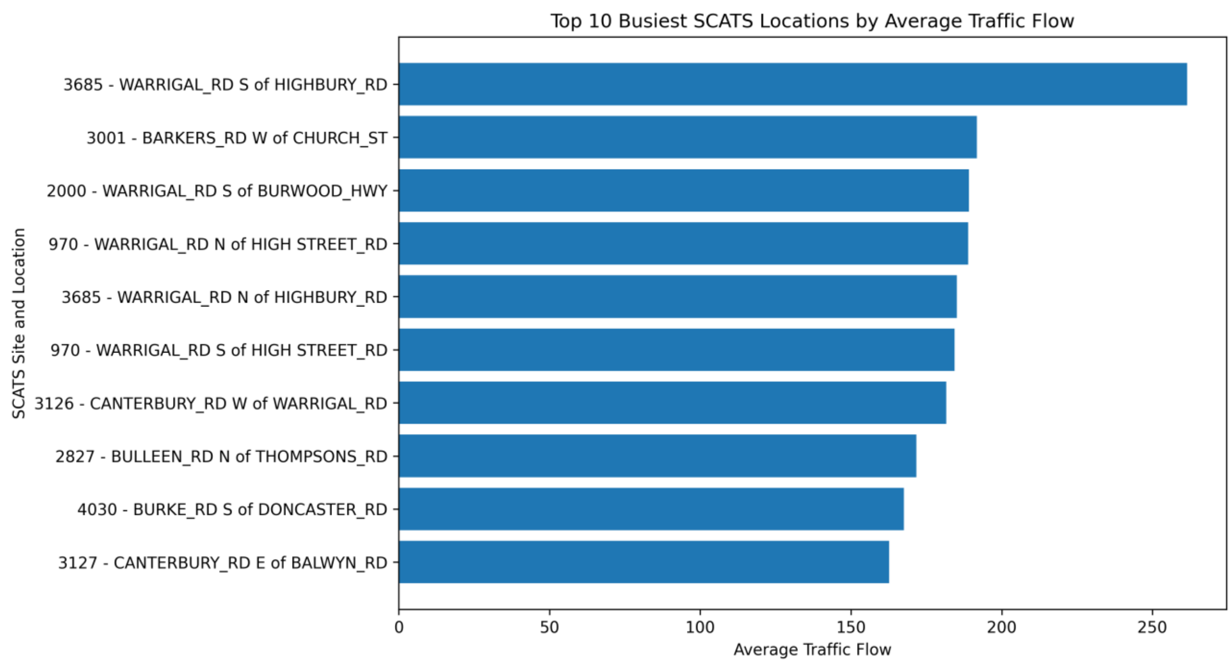

Figure 3. Top 10 recorded SCATS locations by average traffic flow.

The highest-volume location was SCATS site 3685, WARRIGAL RD south of HIGHBURY RD, with an average traffic flow of 261.49. It stands noticeably above the next highest locations. Warrigal Road also appears repeatedly among the higher-volume entries, suggesting that it is a useful corridor for later demonstration or testing of traffic-sensitive route recommendations.

7. Weekday and Weekend Observation

A further comparison was made between weekday and weekend records. Weekday traffic had a higher mean flow of 110.00, while weekend traffic recorded a mean flow of 89.31. The median values show the same pattern: 101 during weekdays compared with 76 during weekends.

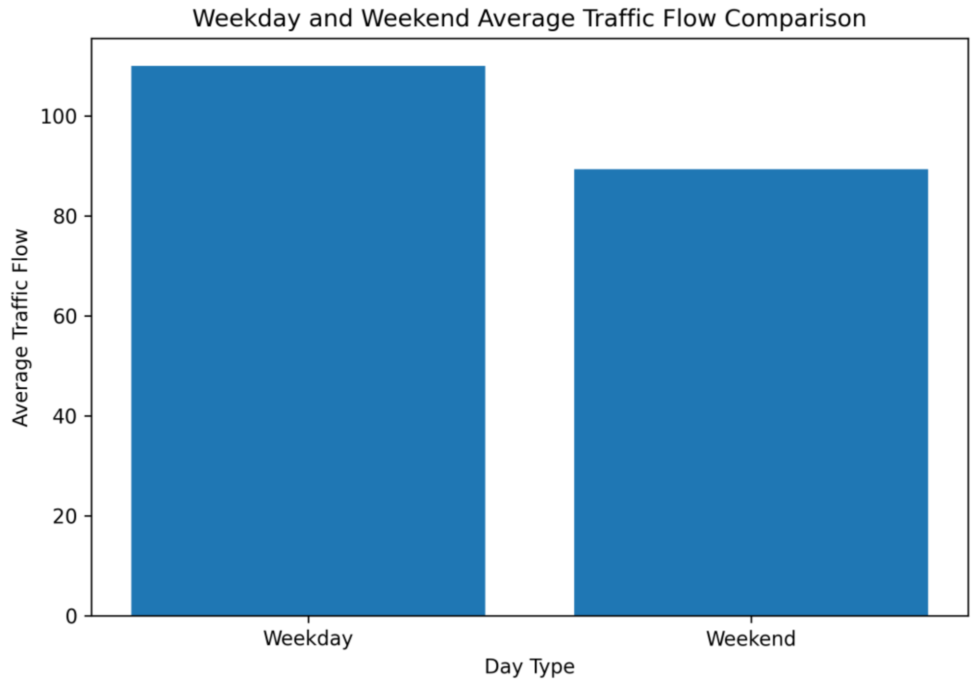

Figure 4. Comparison of mean traffic flow for weekday and weekend records.

This result suggests that traffic behaviour depends not only on time of day, but also on day type. While this investigation does not change the required modelling method, it highlights a factor that may affect prediction quality and could be considered in later model improvement.

8. Relevance to the Traffic-Based Route Guidance System

The research findings provide a clear reason for using traffic prediction within the TBRGS. The data shows substantial variation between quiet and congested periods, and it also shows that some locations consistently carry higher traffic volumes than others. A route-search method that relies only on static distance or a fixed edge cost cannot reflect this variation.

When the group integrates its machine-learning predictions with the route-search component from Part A, the most meaningful demonstration cases should include an evening peak period and at least one busy location identified in this analysis. This would allow the final system to show how predicted traffic conditions influence travel-time estimation and route selection.

9. Limitations

The analysis is based on one month of historical data from October 2006 and a limited geographical area within Boroondara. It should therefore not be treated as a description of current Melbourne traffic. In addition, the dataset does not directly account for incidents, roadworks, weather or special events. These factors could alter traffic behaviour in a real-world routing application.

This research is also intended as a supporting initiative rather than a substitute for the required comparison of LSTM, GRU and the third model. Its contribution is to identify meaningful traffic scenarios and explain why prediction-based travel time is relevant to the end-to-end system.

10. Conclusion

This research investigated peak and off-peak traffic patterns in the supplied Boroondara SCATS dataset. The analysis identified a clear evening traffic peak, with the busiest average 15-minute interval occurring at 5:15 pm. It also found that the traffic volume at the busiest recorded location, SCATS 3685 on Warrigal Road south of Highbury Road, was substantially higher than many other locations in the dataset. Weekday traffic was also higher on average than weekend traffic.

Together, these findings demonstrate that traffic conditions vary meaningfully across both time and location. This supports the purpose of the TBRGS: route recommendations should use predicted traffic and estimated travel time rather than treating every road connection as having a fixed cost.

Resources Used for This Research Section

Table 5. Research resources

Resource	How it was used
Scats Data October 2006.xls	Supplied dataset used to calculate traffic summaries, peak/off-peak periods, location comparisons and weekday/weekend observations.
COS30019 Assignment 2 Part B Brief	Used to ensure that the research initiative is relevant to the required Traffic-Based Route Guidance System and its integration objectives.

References

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